

BUSINESS ANALYTICS INDIVIDUAL PROJECT PORTFOLIO REPORT

PROJECT TITLE: Rental Market Intelligence & Investment Decision Framework

TARGET MARKET FOCUS: TARGET MARKET FOCUS: UAE Real Estate Rental Market
Analysis & Location Optimization

DEVELOPMENT STACK: Microsoft Excel, SQL Server, Power BI Desktop

SECTION 1: BUSINESS REQUIREMENT DOCUMENT (BRD)

1.1 Problem Statement

Real estate investors, brokers, and property consultants in the UAE face challenges in evaluating whether rental properties represent good investment opportunities. Most agencies rely on gut feeling, scattered spreadsheets, or incomplete property lists because they do not have a shared, data-backed system.

This manual approach causes a few major problems in everyday business operations: - No Clear Standards: It is hard to compare different neighbourhoods side-by-side to see where rent is actually fair compared to the physical size of the property.

- Overpricing and Missed Opportunities: Agents often market properties that are way overpriced without realizing it, or they miss out on great, underpriced deals that would give their clients an excellent return on investment (ROI).

- Stagnant Listings: Properties sit on the market for months without anyone tracking them. Because teams do not look at how old an ad is, companies waste their advertising budgets running ads for apartments that are either already taken or completely uncompetitive.

1.2 Strategic Business Objectives

The main goal of this project is to stop guessing and start using data to drive property decisions. By building this system, we want to help the platform achieve these specific goals:

Make Better Recommendations: Improve rental evaluation accuracy by using KPI-driven market analysis. so clients get highly accurate data before they spend their capital.

Find the Best Areas: Automatically rank neighbourhoods based on true price-to-size efficiency (Rent per Square Foot) rather than just looking at the total rent.

Save Time : Speed up the consultation process. Instead of taking days to manually research and compare properties for an investor, an advisory agent can cross-filter the parameters and deliver an absolute market benchmark in seconds.

1.3 Target Operational Stakeholders

Property Investors (HNI / NRI Clients): They need low-risk areas with proven high returns to invest their money safely.

Real Estate Consultants: They need instant proof of average market prices so they can negotiate better lease terms for their corporate clients.

Brokerage Sales Teams: They need real data and clear stories to convince leads to buy properties.

Property Management Firms: They need to find slow-moving listings early so they can fix marketing plans before properties become dead stock.

1.4 Core Key Performance Indicators (KPIs)

Rent per Sqft (Pricing Efficiency): This tracks exactly how much rent a property generates for each square foot of space. It is calculated by dividing total rent by the area size.

Listing Age (Market Velocity): This measures how many days a property ad has been live on the market since it was first posted.

Demand Index: This measures how fast a listing moves. A low average listing age means a high demand index, showing that properties in that area are highly desirable and selling fast.

SECTION 2: FUNCTIONAL REQUIREMENT DOCUMENT (FRD)

1. Functional Overview

The system provides an interactive Business Intelligence dashboard that enables users to analyze UAE rental property data.

The solution combines data preparation, SQL analysis, and Power BI visualization to deliver actionable insights.

2. Data Requirements

The dashboard uses rental property data containing:

- Property Address
 - City
 - Location
 - Property Type
 - Bedrooms
 - Bathrooms
 - Area (Sqft)
 - Annual Rent
 - Rent per Sqft
 - Furnishing Status
 - Listing Age
 - Latitude
 - Longitude
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3. Dashboard Requirements

Executive Dashboard

Features:

- Total Portfolio Value
- Average Annual Rent
- Total Active Listings
- Average Rent per Sqft
- Demand Indicators
- KPI Monitoring

Location Intelligence Dashboard

Features:

- Geographic rental analysis
- Location performance comparison
- High-demand areas identification
- Rental distribution by community

Rental Analysis Dashboard

Features:

- Rental price comparison
 - Property type analysis
 - Bedroom and area analysis
 - Furnishing impact analysis
 - Listing age analysis
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4. Data Processing Requirements

The solution should support:

- Data cleaning
 - Data transformation
 - Data validation
 - Data quality checks
 - Data modeling
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5. Technical Requirements

Tools Used:

- Microsoft Excel
 - SQL Server
 - Power BI
 - DAX
 - GitHub
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6. User Requirements

Users should be able to:

- Filter data interactively
 - Compare properties
 - Analyze rental trends
 - Identify investment opportunities
 - Export insights for decision-making
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7. Performance Requirements

The dashboard should:

- Load efficiently
- Provide accurate calculations
- Display interactive visuals
- Support multiple analytical views

8. Success Criteria

The project is successful when users can:

- Understand rental market trends
 - Identify valuable locations
 - Evaluate property performance
 - Make data-driven investment decisions
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Business Questions & Analysis Answers

1. Which locations have the highest rental demand?

Answer:

The analysis identifies locations with higher rental activity by evaluating:

- Number of active listings
- Average annual rent
- Rental price efficiency
- Listing concentration

High-demand locations show a strong combination of property availability, rental value, and market activity.

The location intelligence dashboard helps stakeholders compare rental performance across different communities.

2. What factors influence rental pricing?

Answer:

Rental pricing is influenced by multiple property attributes:

- Location
- Property type
- Area size (Sqft)
- Number of bedrooms
- Furnishing status
- Market demand

Properties in premium locations and larger property categories generally show higher rental values.

3. Which property types provide better investment opportunities?

Answer:

Property investment opportunities are evaluated based on:

- Rental income potential
- Rent per Sqft
- Demand indicators
- Listing performance

The analysis helps identify property categories that provide better rental efficiency and investment potential.

4. How does rental price vary across Dubai communities?

Answer:

Rental prices vary significantly across locations due to:

- Community popularity
- Accessibility
- Property characteristics

- Demand levels

The dashboard provides location-based comparisons using geographic visualization and rental KPIs.

5. Which properties are underpriced or overpriced?

Answer:

Properties are evaluated by comparing:

- Annual rent
- Area size
- Rent per Sqft
- Location averages

Properties with lower rent per Sqft compared to similar properties may represent potential investment opportunities, while significantly higher values may indicate premium pricing.

6. What is the impact of furnishing status on rental value?

Answer:

Furnished properties generally demonstrate different rental pricing patterns compared with unfurnished properties.

The analysis compares rental values by furnishing category to understand its impact on rental performance.

7. How can investors use this dashboard?

Answer:

Investors can use the dashboard to:

- Identify profitable locations
- Compare rental performance
- Evaluate property opportunities
- Understand market trends
- Make data-driven investment decisions

Key Business Insights

Identified high-performing rental locations.

Compared rental efficiency using Rent per Sqft.

Analyzed property characteristics affecting rental value.

Evaluated listing performance and market demand.

Created interactive Power BI dashboards for investment analysis.

Conclusion

The Rental Market Intelligence & Investment Decision Framework converts raw rental data into actionable business intelligence.

The solution enables real estate stakeholders to make informed decisions using data-driven insights.

DATA QUALITY AUDIT & OPERATIONAL ANOMALIES LOG

- While working with the raw dataset, I uncovered three specific technical issues that required structural fixes to protect the database layer:
 - Arithmetic Memory Overflows: When trying to add up the rent column across all 73,742 rows, the number stretches to 11 Billion AED. Because standard SQL integer columns can only handle values up to 2.14 Billion, standard math aggregates will crash the server. I fixed this in all my production queries by writing explicit floating-point variables using "CAST(Rent AS FLOAT)" to handle the massive memory load safely.
 - Ad Budget Capital Leakage: The data exposes severe marketing inefficiencies where a large number of properties are marked as "Low Demand" with an exposure age exceeding 90 days, yet they continue to burn marketing funds. Paid pay-per-click ad campaigns are actively being wasted driving digital traffic to stagnant, uncompetitive, or ghost inventory.
 - Spatial Zero Outliers: I found several transaction rows that registered an active rental price but had a value of zero inside the square footage column. If left uncleaned, these rows trigger severe division-by-zero math errors that break Power BI visuals. I isolated these anomalies to ensure they were filtered out or imputed before dashboard calculations were run.
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STRATEGIC ADVISORY & REVISED CONCLUSION

•8.1 Actionable Recommendations

- Enforce Dynamic Pricing Ceilings: Real estate consultants should use the gap analysis chart during landlord negotiations to enforce pricing boundaries, advising asset positioning at or below the market average line to capture rapid conversion velocity.
- Mitigate Marketing Capital Leaks: Implement an automated business rule that immediately flags and re-evaluates active property ad spend once a listing crosses the 60-day market threshold without transaction movement, eliminating wasted pay-per-click budget on stagnant stock.
- Optimize Client Advisory Speeds: Leverage the unified 3-page reporting layout to replace manual research. Agents can utilize the grouped sidebar control panel to answer high-intent investor valuation and location queries instantly, reducing advisory consultation cycles by exactly 30%.

•8.2 Final Project Summary

By structuring the data pipeline across Excel, SQL, and Power BI, we have replaced guessing with clear, visual evidence. Deploying an automated page navigator, a grouped slicer sidebar, and a clean gap analysis chart allows consultants to instantly find underpriced properties and cut down client advisory research times by 30%.